

HDOC - Day 38

String Operations Using Separate Functions

Algorithm

1. Display the menu repeatedly. 2. Option 1 reads a string and counts spaces, digits and special characters by scanning each character. 3. Option 2 reads a string and a given character and counts its occurrences. 4. Option 3 toggles every alphabetic character: uppercase becomes lowercase and lowercase becomes uppercase; other characters remain unchanged. 5. Exit on option 4 and show an error for an invalid choice.

Test Case Table

T C	Input	Expected	Actual	Result
1	Hello 123! C; op 1	Spaces=2, Digits=3, Special=1	2,3,1	Pass
2	programming, m; op 2	Frequency=2	Frequency=2	Pass
3	Hello CSE 123!; op 3	hELLO cse 123!	hELLO cse 123!	Pass

C Program

```
#include <stdio.h>
void countAll(void); void frequency(void); void toggle(void);
int main(){
    int ch;
    do{
        printf("\n1. Count spaces, digits and special characters\n2. Character frequency\n3. Toggle case\n4. Exit\nEnter choice: ");
        scanf("%d",&ch); getchar();
        switch(ch){case 1:countAll();break;case 2:frequency();break;case 3:toggle();break;
                    case 4:break;default:printf("Invalid choice.\n");}
    }while(ch!=4); return 0;
}
void countAll(void){
    char a[200]; int i,sp=0,dg=0,sc=0;
    printf("Enter string: "); fgets(a,200,stdin);
    for(i=0;a[i] && a[i]!='\n';i++){
        if(a[i]==' ') sp++;
        else if(a[i]>='0'&&a[i]<='9') dg++;
        else if(!((a[i]>='A'&&a[i]<='Z')||((a[i]>='a'&&a[i]<='z')))) sc++;
    }
    printf("Spaces = %d\nDigits = %d\nSpecial characters = %d\n",sp,dg,sc);
}
void frequency(void){
    char a[200],ch; int i,c=0;
    printf("Enter string: "); fgets(a,200,stdin);
    printf("Enter character: "); scanf("%c",&ch);
    for(i=0;a[i]&&a[i]!='\n';i++) if(a[i]==ch)c++;
    printf("Frequency of '%c' = %d\n",ch,c);
}
void toggle(void){
    char a[200]; int i;
    printf("Enter string: "); fgets(a,200,stdin);
    for(i=0;a[i]&&a[i]!='\n';i++){
        if(a[i]>='A'&&a[i]<='Z') a[i]+=32;
        else if(a[i]>='a'&&a[i]<='z') a[i]-=32;
    }
    printf("Toggled string: %s",a);
}
```

Compilation, Execution & Testing

```
puppie@Anvikas-MacBook-Air Desktop % clang day38.c -o day38
puppie@Anvikas-MacBook-Air Desktop % ./day38
Enter choice: 1
Enter string: Hello 123! C
Spaces = 2
Digits = 3
Special characters = 1
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day38
Enter choice: 2
Enter string: programming
Enter character: m
Frequency of 'm' = 2
```

```
puppie@Anvikas-MacBook-Air Desktop % ./day38
Enter choice: 3
Enter string: Hello CSE 123!
Toggled string: hELLO cse 123!
```

Result: The menu-driven string program was compiled, executed and tested successfully for all three operations.